

Technical Approach

Expanded technical architecture and data/model explanation for Slide 3. The supplied slide specifies the GitHub repository and the major frontend, backend, database, AI/remote-sensing and DevOps components. [\[filecite\]turn1file0\[L24-L28\]](#)

1. End-to-end architecture

Layer 1 — Data ingestion: satellite/weather, geological records, exploration reports, assays, production and equipment information.

Layer 2 — Data engineering: coordinate normalization, timestamp alignment, missing-value handling, feature generation and geospatial indexing.

Layer 3 — Intelligence: spatial modelling, anomaly detection, time-series forecasting, optimization and rule-based constraints.

Layer 4 — Decision services: APIs expose predictions, risks and optimization results to the frontend.

Layer 5 — Geospatial UI: mine maps, 3D blocks, risk layers, scenario outputs and evidence/audit views.

2. Satellite and environmental data

MOSDAC's GSMap_ISRO Rain is an especially relevant environmental source. ISRO describes it as a gauge-adjusted satellite rainfall product with $0.1^\circ \times 0.1^\circ$ spatial resolution, 1-hour temporal resolution, coverage from March 2000 onward and open access. [\[cite\]turn0search0](#)

Sentinel-1 adds C-band SAR. ESA states that Sentinel-1 can image Earth through cloud and rain and regardless of day/night, and that its land applications include monitoring land-use changes and ground movement. [\[cite\]turn0search3\[turn0search14\]](#)

For NAKSHATRA-X, these sources are best used for environmental state and disruption monitoring, not as a standalone deep-ore detector.

3. Geological data and reserve modelling

GSI's mineral-hunt portal exposes dedicated routes for Geological Data, Exploration Reports and Data, and Remote Sensing Data. [\[cite\]turn0search11](#)

Use borehole collars, depth intervals, assay values and geological domains to create a spatial training dataset. A Kriging-based model can interpolate grades where assumptions about spatial continuity are reasonable.

Validation should be spatially separated. Randomly splitting nearby boreholes can create leakage because neighbouring samples are correlated. A stronger test holds out spatial blocks or drill sections.

4. Machine-learning layer

The slide proposes probabilistic/ensemble reserve mapping, time-series forecasting and a rule-based + ML hybrid recommendation engine. The hybrid design is important because mining decisions have hard constraints that a purely statistical model may violate.

Recommended ML outputs should include prediction + confidence/uncertainty + explanation + data freshness.

Forecast models should use rolling time splits rather than random train/test splitting. For each forecast, report MAE/MAPE where appropriate, prediction intervals and performance by operating regime.

5. Optimization layer

For ore blending, formulate a constrained optimization problem: choose quantities from stockpiles subject to tonnage, grade and availability constraints while minimizing cost or deviation from the desired blend.

SciPy's linprog provides a standard linear-programming formulation with bounds and equality/inequality constraints. The optimization output should show the active constraints and sensitivity to stockpile assays.

Do not describe the optimizer as "guaranteeing" a grade unless the input assays and process assumptions support that guarantee.

6. Backend and deployment

The supplied architecture lists FastAPI/Python, PostgreSQL/PostGIS, React/Mapbox, cloud GPU inference, Docker and Kubernetes. The repository listed on the original slide is <https://github.com/29rdx/newNakshatra-x>.
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For the national-level prototype, demonstrate one API end-to-end rather than listing every technology. The judge should be able to see input → computation → output → decision.

7. Security and auditability

Every model result should carry a model version, data timestamp, input dataset identifier and recommendation ID. For high-impact actions, store the recommendation and approval state.

If cryptographic hashing is used, demonstrate one real audit record and its verification rather than only describing SHA-256 in the architecture.

Evidence and exact source paths

ID	Source	Exact resource / URL	Use in NAKSHATRA-X
S1	MOIL 63rd Annual Report 2024-25	https://moil.nic.in/userfiles/Annual_Report_2024_25.pdf	Reserves/resources, exploration drilling, production and strategic target.
S2	MOIL MD&A; / Annual Report copy	https://moil.nic.in/userfiles/file/InvRel/Financials/Annual_Report_2024-25.pdf	FY25 manganese production and import context.
S3	MOSDAC — GSMap ISRO Rain	https://www.mosdac.gov.in/gsmmap-isro-rain	Exact rainfall product; hourly, 0.1° and March 2000 onward.
S4	GSI Reports database	https://gsi.gov.in/gsiReport/	Official geological/exploration report search.
S5	GSI Mineral Hunt data portal	https://hackathon.gsi.gov.in/login	Direct geology, exploration and remote-sensing data routes.
S6	GSI geology data	https://nis.bisag-n.gov.in/download/Geology/	Direct geology data directory.
S7	GSI exploration reports/data	https://nis.bisag-n.gov.in/download/Exploration_data_reports/	Direct exploration data/report directory.
S8	GSI remote sensing ASTER	https://nis.bisag-n.gov.in/download/Remotesensing_ASTER/	Direct remote-sensing directory.
S9	GSI/OGD manganese deposit dataset	https://tn.data.gov.in/catalog/location-manganese-ore-deposits-in-dia-and-its-salient-features	Specific manganese-deposit location/geology dataset.
S10	Indian Minerals Yearbook portal	https://ibm.gov.in/IBMPortal/pages/Indian_Minerals_Yearbook	Official IBM source for manganese statistics.
S11	Indian Minerals Yearbook 2024	https://ibm.gov.in/writereaddata/files/177426215469c1178a48453IMYB_2024_EBookFinal.pdf	Producer/location and manganese industry data.
S12	ESA — Sentinel-1 seeing through clouds	https://www.esa.int/ESA_Multimedia/Videos/2014/03/Sentinel-1_seeing_through_clouds	SAR capability through cloud/rain and day/night.
S13	ESA — Introducing Sentinel-1	https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1	C-band SAR, all-weather/day-night mission context.
S14	SciPy linprog API reference	https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html	Linear-programming solver used for constrained blending formulations.
S15	Team GitHub repository	https://github.com/29rdx/newNakshatra-x	Implementation repository named in the submitted technical slide.

Recommended technical benchmark table

Reserve model: MAE/RMSE, bias, uncertainty coverage and spatial hold-out performance.

Risk model: precision, recall, false-alert rate and warning lead time.

Forecast: MAE/MAPE, prediction interval coverage and rolling-origin validation.

Optimizer: grade deviation, tonnage feasibility, constraint violations and runtime.

System: API p50/p95 latency, data freshness, offline fallback time and audit completeness.